29 The table shows how the count rate from a sample of radioactive nuclide varies with time *t*. The nuclide has a half life of 50 s. The average background count rate is constant.

t / s	Count rate / Bq
0	72
50	40

What count rate will be measured when $t = 150$ s?

A 8 Bq **B** 16 Bq **C** 24 Bq **D** 32 Bq